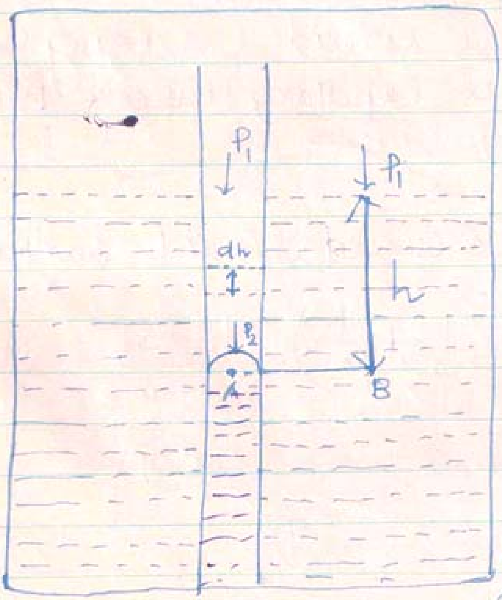
Surface Tension and Evaporation:

When a liquid drop evaporates under normal temp; the latent heat required for the change in state is partially supplied from the free surface energy, which is released; as the surface area of the drop decreases.

Thus a liquid drop continues to evaporate in a space which is in vapour equilibrium with the flat surface of that liquid.

We now show how the equilibrium state of saturated vapour and its liquid depends on the curvature of the liquid surface.



Let us consider a closed space containing a liquid and its saturated vapour in equilibrium.

Let a capillary tube (shown magnified in the fig.) whose wall is not wetted by the liquid be dipped in the liquid. The liquid level inside the capillary tube falls below the level outside the capillary tube.

S = Surface tension of the liquid.
ρ = Density of the liquid.
r = radius of the miniscus (curved liquid surface)
h = The depression of the liquid level inside the capillary tube.

let P_1 = S.V.P over the flat liquid surface
P_2 = The S.V.P over the curved liquid surface.

Let us consider a vapour column of height 'dh' inside the capillary tube.
σ = The density of that vapour column.
Pressure exerted by that vapour column of height dh

$$dP = dh\,\sigma g \longrightarrow (1)$$

$$\therefore P_2 = P_1 + \int dh\,\sigma g \longrightarrow (2)$$

$\int dh\,\sigma g$ = The total Pre? exerted by the vapour column of height 'h'.
Let us consider two points A and B on the same level inside the liquid; A being just below the miniscus in the capillary tube & B is outside the capillary tube.

$$P_A = P_B$$

The excess pressure on the curved liquid surface

$$P_A - P_2 = S\left(\frac{1}{r} + \frac{1}{r}\right) = \frac{2S}{r}$$

$$\text{or } P_A = P_2 + \frac{2S}{r} \longrightarrow (3)$$

$$P_B = P_1 + h\rho g \longrightarrow (4)$$

Equating (3) and (4) :-

$$P_2 + \frac{2S}{r} = P_1 + h\rho g$$

$$\frac{2S}{r} = h\rho g - (P_2 - P_1)$$

$$\text{or, } \frac{2S}{r} = \int dh\rho g - \int dh\,\sigma g$$

$$\text{or } \frac{2S}{r} = \int (\rho - \sigma)\, dh\, g = \int \frac{(\rho - \sigma)}{\sigma}\, dh\,\sigma g$$

Or, $\frac{2S}{r} = \int_{P_1}^{P_2} \frac{(\rho-\sigma)}{\sigma} . dP$

Since the density of saturated vapour is very much less than the density of that liquid, $\sigma << \rho$ hence σ can be neglected compared to ρ.

So, we have $\frac{2S}{r} = \int_{P_1}^{P_2} \frac{\rho}{\sigma} dP$ ———— (5)

Let us consider a volume 'V' of the saturated vapour at pressure P. Let the density of that vapour be σ. Since saturated vapour pressure obeys gas law, we have from the gas equation;

$$PV = nRT$$

$$P . \frac{m}{\sigma} = \frac{m}{M} R.T$$

$$\frac{1}{\sigma} = \frac{1}{P} . \frac{RT}{M} \quad \text{———— (6)}$$

Where M = Molecular weight of the liquid.
T = Temp. of the liquid.

Putting equation no. 6 in 5

$$\frac{2S}{r} = \int_{P_1}^{P_2} \rho \frac{RT}{M} \frac{dP}{P}$$

$$= \frac{RT\rho}{M} \int_{P_1}^{P_2} \frac{dP}{P} = \frac{RT\rho}{M} [\log P]_{P_1}^{P_2}$$

$$= \frac{RT\rho}{M} [\log_e P_2 - \log_e P_1] = \frac{RT\rho}{M} \log_e \frac{P_2}{P_1}$$

$$\therefore \boxed{\log_e \frac{P_2}{P_1} = \frac{2S.M}{RT\rho r}} \quad \text{———— (7)}$$

Equation (7) gives a relation between Saturated vapour pressure over the flat surface (P_1), that of over the curved surface (P_2) and radius of curvature.
Since S, M, R, ρ, r and T are all positive quantities ; the R.H.S of equation no.(7) is positive

Hence $\log_e \frac{P_2}{P_1} = +ve$ i.e $P_2 > P_1$

Hence the Saturated vapour pressure over curved surface is greater than that of over the flat surface.
At 0°C (T = 273K) for water the saturated vapour pressure over the flat surface $P_1 = 0.46$ cm of Hg
for water $S \simeq 70$ dynes/cm, M = 18, ρ = 1gm/c.c
T = 273K, R = 8.3×10^7 ergs / °c / mol
Using equation (7) we can find the $\log_e \frac{P_2}{P_1}$ for drops of different radii and the result are given below
For water at 0°C

Radius of the drop 'r'	$\log_e \frac{P_2}{P_1}$	P_2
50×10^{-7} cm	1.02	0.47 cm of Hg
5×10^{-7} cm	1.26	0.58 cm of Hg
$.5 \times 10^{-7}$ cm	10.2	4.7 cm of Hg

With the help of this table, we now analyze our theoritical result.
Let us consider a closed space containing water

vapour only at $0^\circ C$ and having a vapour pressure of 0.58 cm of Hg. The space is large enough so that a little evaporation into that space or a little condensation does not change the vapour pressure.

0.58 cm of Hg

$r = 5\times10^{-7}$ cm → Space is just saturated w.r to this drop. Hence the drop remains in equilibrium in that space.
($P_2 = .58$ cm Hg)

$r = .5\times10^{-7}$ cm → Space is unsaturated w.r to this drop. The drop evaporates in that space.
($P_2 = 4.7$)

$r = 50\times10^{-7}$ cm → space is super saturated w.r to this drop the vapour condenses the drop grows in size.
($P_2 = .47$)

Let us consider a water drop of radius $r = 5\times10^{-7}$ cm in that space. From the table, we find that the saturated vapour pressure for water drop of that radius is 0.58 cm of Hg. Since the vapour pressure in that space is also 0.58 cm of Hg; the space is just saturated w.r to that drop. Hence the drop remains in equilibrium in that space.

Let us consider a drop of radius $r = .5\times10^{-7}$ cm in that space; and the saturated vapour pressure for that drop is 4.7 cm of Hg as seen from the table. The vapour pressure in that space being much less than the S.V.P; the space is unsaturated w.r to that drop. Hence the drop

Continues to evaporate in that space.

Let us consider a drop of radius $r = 50 \times 10^{-7}$ cm for which S.V.P as seen from the table is 0.47 cm of Hg. The vapour pressure in that space being greater than the S.V.P of the drop; the space is super saturated w.r to that drop; and hence condensation takes place and the drop grows in size.

Thus in a given space, containing a given amount of water vapour; drops of a particular radius can only remain in equilibrium. Drops of radii less than that particular value will evaporate in that space, while drops of radii greater than that particular value will continue to grow in size.

Nuclei for Condensation:

From the above analysis we find that; in a space, containing vapour only; condensation will never take place, whatever be the amount of vapour present; unless the space finds a drop with respect to which the space can be regarded as super saturated. Thus in a given space, to start condensation, we require a drop around which condensation will take place and is known as nucleus for condensation. Generally, the dust particles in atmosphere serve as nuclei for condensation.

SURFACE TENSION AND BOILING

For normal boiling both gases and vapour must be present inside the bubble.

let P_1 = Pressure of the vapour present in the bubble.
P_2 = Pressure of the gas present in the bubble.
r = Radius of the bubble.

We assume that, the bubble is just near the surface.

P_A = The atmospheric pressure.
S = Surface tension of the liquid.

The excess pressure inside the bubble ;

$$P = (P_1 + P_2) - P_A = \frac{2S}{r}$$

$$(P_1 - P_A) + P_2 = \frac{2S}{r}$$

$$\therefore \boxed{P_1 - P_A = \frac{2S}{r} - P_2} \text{ ------ } (1)$$

At the time of boiling, the ~~atmospheric~~ temperature of the liquid remains constant and hence the gas present inside the bubble will obey Boyle's law.

$$P_2 V = \text{Constant} = K$$

$$\text{or } P_2 = \frac{K}{\frac{4}{3}\pi r^3} \qquad \left[\because \text{volume of the bubble} = \frac{4}{3}\pi r^3\right]$$

$$\therefore P_2 = \frac{K'}{r^3} \text{ ------ } (2) \quad \text{Where } K' = \frac{3K}{4\pi} = \text{Const.}$$

Putting eqn (2) in (1) :-

$$P_1 - P_A = \frac{2S}{r} - \frac{K'}{r^3} \text{ ------ } (3)$$

The K' is an unknown constant.

We know that when a liquid boils ; the vapour pressure P_1 remains constant. Since P_A is also constant so $(P_1 - P_A)$ is constant, but the radius of the bubble changes.

On differentiating L.H.S of eqn (3) w.r to r :

$$\frac{d}{dr}(P_1 - P_A) = 0$$

Now differentiating eqn (3) w.r to 'r' and equating it to zero.

$$-\frac{2S}{r^2} - \left(-\frac{3}{r^4}\right)K' = 0$$

$$\text{or,} \quad \frac{3K'}{r^4 r^2} = \frac{2S}{r^2}$$

$$\therefore K' = \frac{2}{3} S r^2 \quad \text{———— (4)}$$

Putting equation (4) in (3) :-

$$P_1 - P_A = \frac{2S}{r} - \frac{2Sr^2}{3r^3}$$

$$\therefore P_1 - P_A = \frac{2S}{r} - \frac{2S}{3r} = \frac{4S}{3r}$$

$$\therefore \boxed{P_1 - P_A = \frac{4S}{3r}} \quad \text{———— (5)}$$

From eqn (5) we find that, a bubble of radius 'r' will boil only when the vapour pressure inside the bubble exceeds the atmospheric pressure by $\frac{4S}{3r}$.

VARIATION OF SURFACE TENSION WITH TEMP.

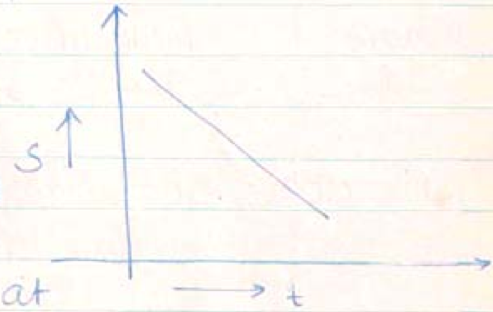
For a small range of temperature, the variation of surface tension with temp. is given by

$$S_t = S_0(1-\alpha t)$$

Where S_0 and S_t = Surface tension of the liquid at $0^\circ C$ and $t^\circ C$ respectively.

α = Constant = Temp. Coefficient of surface tension.

$$\frac{dS_t}{dt} = -S_0\alpha = \text{Constant}$$



Since at critical temp. T_c, no surface exists between liquid state and vapour state, hence surface tension must vanish at critical temp. T_c.

Vanderwall's suggested formula on the variation of surface tension with temperature:

Vanderwall's suggested formula:

$$S = A\left[1-\frac{T}{T_c}\right]^{\frac{3}{2}}$$

Later it was found that, a better result can be obtained for most of the liquids by using the relation;

$$S = A\left[1-\frac{T}{T_c}\right]^{n}$$

Where $n = 1.21$

$\therefore \boxed{\frac{dS}{dT} = -\frac{A}{T_c}}$ → Thus for liquids $\frac{dS}{dT}$ is constant.

~~Eotov~~ Eotvos Law :-

We know that surface area $\propto$ (volume)$^{2/3}$

How? $A \propto r^2$ $\quad r \propto A^{1/2}$

$V = Kr^3 \quad V = A^{3/2} \quad \therefore A = V^{2/3}$

Let us consider 1 gm-mole of the liquid volume of 1 gm-mol of the liquid i.e the molar volume $= \left(\frac{M}{\rho}\right)^{2/3}$?

Where M = Molecular weight of the liquid & ρ = density of the liquid.

$\therefore$ The molar surface area $= K'$ (molar volume)$^{2/3}$

$$= K'\left(\frac{M}{\rho}\right)^{\frac{2}{3}}$$

$\therefore$ The molar surface energy of the liquid

$$= S \cdot K'\left(\frac{M}{\rho}\right)^{\frac{2}{3}}$$

($\because$ Surface energy = Surface tension × change in surface area).

Eotvos law states that, the molar surface energy of a liquid is proportional to $T_c - T$

i.e $\quad S \cdot K'\left(\frac{M}{\rho}\right)^{\frac{2}{3}} = K''(T_c - T)$

$$\therefore S\left(\frac{M}{\rho}\right)^{\frac{2}{3}} = \frac{K''}{K'}(T_c - T)$$

$$\therefore \boxed{S\left(\frac{M}{\rho}\right)^{\frac{2}{3}} = K(T_c - T)} \quad \text{——— (1)}$$

Where K is a new constant.

Ramsay and Shield gave a modified formula

$$S\left(\frac{M}{\rho}\right)^{\frac{2}{3}} = K\left(T_c - \delta - T\right) \text{ ------ } (2)$$

where $(T_c - \delta)$ is the temp. at which the liquid surface vanishes.

Katayama's Relation :

$$S\left(\frac{M}{\rho-\rho'}\right)^{\frac{2}{3}} = a\left(T_c - T\right) \text{ ——— } (3)$$

where ρ = density of the liquid.
ρ' = density of the saturated vapour.
a = constant.

We know that, free surface energy is given by

$$\boxed{E = S - T\frac{dS}{dT}}$$

[Can be proved → Newman & Searle].

Differentiating w.r to T

$$\frac{dE}{dT} = \cancel{\frac{dS}{dT}} - T\frac{d^2S}{dT^2} - \cancel{\frac{dS}{dT}}$$

$$\therefore \frac{dE}{dT} = -T\frac{d^2S}{dT^2}$$

But since $\frac{dS}{dT}$ is approximately constant for a given liquid, so $\frac{d^2S}{dT^2} = 0$

$$\therefore \frac{dE}{dT} = 0$$

$$\therefore E = \text{Constant}$$

Thus the free surface energy of a liquid is constant and does not change with temperature.